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Experimental Evaluation of a Campus-deployed IoT Network using LoRa

Parag Kulkarni, Qornitah Abdul Hakim, Abderrahmane Lakas

***Abstract*—The emergence of the Internet of Things (IoT) has redefined sensor networking with an unprecedented increase in the number of applications leveraging sensor technology in a myriad of domains. This has also revolutionized the communication technology required to network these sensors to exploit their potential. Short range wireless technologies provide high data rate and typically operate in the unlicensed band. Cellular technologies on the other hand provide relatively longer range, higher data rate and operate in the licensed spectrum. The need to bridge the gap between these two, i.e., provide a low cost and longer range solution with the ability to connect massive number of geographically spread out devices using the unlicensed spectrum led to the emergence of Low Power Wide Area Networks (LPWAN). Long Range (LoRa), one of the LPWAN technologies continues to attract attention given its open specification and ready availability of off-the-shelf hardware with claims of several kms of range in harsh challenging environments. In this paper we put these claims under test through extensive measurements carried out over a test-bed deployed in a University campus. The conducted tests focused on the link quality and transmission performance based on various parameters such as the spreading factor (SF), coding rates (CR), bandwidth for different radio propagation scenarios. Findings from this study show that the achievable performance can greatly vary and prompts caution when taking numbers at face value as this can have implications for the IoT applications.**

***Index Terms*—Internet of Things (IoT), Sensor Networks, LoRa, LPWAN**

I. Introduction

INNOVATIONS in wireless networking have all along been driven by the need to enhance spectrum utilisation, improve achievable range and data rate and reduce power consumption at the same time aiming to minimise the cost. Short range wireless technologies such as those embodied in the IEEE 802.11 (WiFi), IEEE 802.15.1 (Bluetooth) and IEEE 802.15.4 typically operate in the Industrial, Scientific and Medical (ISM) bands which are unlicensed bands and provide a high data rate service with relatively smaller coverage in comparison to their cellular counterparts. The cellular communication technologies such as those embodied in the 3G, 4G and the upcoming 5G networks, on the other hand typically operate in the licensed bands (although LTE-U leverages the unlicensed bands too) and provide variety of services catering to different class of applications at a relatively higher cost.

All authors are affiliated to the Computer & Network Engineering Department, College of IT, UAE University, Al Ain, UAE. E-Mail: {parag, 201540260, alakas}@uaeu.ac.ae

Breakthroughs in technology, commoditization and ubiquitous connectivity led to the emergence of the Internet of Things (IoT) paradigm. IoT typically involves deploying sensors reporting data over a communication link to facilitate monitoring/management applications in a variety of domains. One of the key requirements is that devices consume very little power and continue to operate on battery power for years together. Moreover, not all IoT applications require a high data rate service. In fact, a large number of IoT applications are delay and loss tolerant and only transmit once in a while. Most of these applications require low cost connectivity at scale which motivated the need for connectivity alternatives to bridge the gap between the short range and cellular based alternatives. Consequently, Low Power Wide Area Networks (LPWANs) emerged to bridge this gap. An overview of LPWANs can be found in [1]. Although cellular standards have continued to evolve (such as Long Term Evolution Machine type communications (LTE-M), Narrow Band IoT (NB-IoT) and the upcoming 5G massive Machine Type Communications (mMTC) and ultra reliable MTC (uMTC)) to cater to the needs of IoT applications, a forecast from Machina Research [2] estimates that 11% of the 27 Billion global IoT connections by the year 2025 will be served by LPWAN technologies.

Whilst there are many players in the LPWAN space such as Sigfox [3], Ingenu [4] [5], Weightless [6], Long Range (LoRa) [7], the latter [7], a technology currently being offered by SemTech and championed by the LoRa Alliance [8] is gaining popularity. We refer the reader interested in the details of these technologies to [3] [4] [5] [6] [7]. LoRa continues to receive attention from the industrial and academic community due to the openness[1] of its specification and ready availability of hardware in the market which can facilitate experimental research and rapid prototyping. The objective of this work (also the contribution of this paper) has been to study the coverage and performance of this technology, observe and report on the impact of different transmission parameter settings on the performance in different scenarios and highlight the reality, that achieveable performance can vary greatly prompting caution when taking bold claims in the literature at face value.

The rest of the paper is organised as follows: Section II reviews relevant related work, section III provides an

[1]Although communication with off-the-shelf modules is easy and accessible, doing the same with a software radio is impossible. Whilst the physical layer is known and can be demodulated, the packet cannot be decrypted as the details of the encryption are not disclosed.

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overview of the LoRa technology, section IV highlights the methodology and experimental setup used in this study following which the findings from this study are highlighted in section V. Discussion on practical deployment challenges follows in section VI with the paper finally concluding in section VII summarising the findings and highlighting directions for future work.

II. Related Work

There have been several studies on LPWAN technologies with the focus primarily being on LoRa/LoRaWAN. This is hardly surprising given the open nature of the LoRa/LoRaWAN specification and availability of off-the-shelf hardware that can be leveraged for experimental research and rapid prototyping. In addition to empirical research, analytical and simulation studies have also been conducted. Studying the scalability aspects of LoRa has been the focus in [9] [10] [11]. Bor et al. [9] designed and released a simulator called LoRaSim for studying the performance of LoRa networks through simulation. This was further extended by Pop et al. in [12] to support the LoRaWAN MAC protocol. Several observations and potential enhancements related to the operational aspects of LoRaWAN networks were outlined in this paper.

In [13], the authors provide a study on the impact of certain PHY parameters on the performance of a LoRa network deployed in a rural tree plantation in the United States. The analysis shows that communication range of LoRa was smaller than the theoretically expected range. The same study shows that unlike the RSSI, the actual data reliability was inconsistent at varying distances and PHY configurations.

In [14], the authors address the scalability issue in terms of the number of end devices per gateway in the case of a single-gateway LoRaWAN deployments. This study uses a simulation model to assess the scalability of a single-gateway by measuring the impact of the interference caused namely by concurrent transmission. The results show that under certain conditions and lower duty cycles such as 1%, as many as 1000 nodes can be deployed causing only 30% of data loss.

The research in [15], provided a comprehensive performance evaluation of LoRaWAN under different urban, sub-urban and rural settings through static and mobile tests using high gain antennas connected to the gateway and end devices. The gateway was deployed on the roof of a 5-storey building with the antenna pointing towards the test area. A coverage running into several kms was observed in these tests. A theoretical model is also included in this study for radio planning and a tool for generating topographic maps.

The study in [16], focuses on the performance analysis of a single receiver in the presence of collisions, and shows that collisions between packets modulated with different SFs can indeed cause packet loss if the interference power received is strong enough. This study also shows that inter-SF collisions are an issue in LoRa networks. Therefore, allocating higher SFs to users far from the gateway might not necessarily improve their link capacity in case of congested networks, thereby, contradicting the common belief that SFs can be considered orthogonal.

Studies in [17] [18] [19] [20] focus on evaluation of LoRaWAN in indoor and outdoor environments. The study described in [18], in particular, found that by deploying a LoRaWAN gateway on a mountain (approx. 500m above sea level), resulted in a coverage area close to 1400 square kms. This study was conducted in Ireland. Another study conducted in Finland by Petajajarvi et al. [17] found 15-30 km of range using LoRaWAN. Cattani et al. [21] studied the impact of parameter settings and environmental conditions on the energy efficiency of LoRa transceivers. They argue that since LPWANs are suitable for outdoor deployments, it is useful to know the impact of external environmental conditions such as temperature and humidity on the network performance. Their findings indicate that the performance is sensitive to fluctuations in temperature. Rathod et al. [22] elborate on performance comparison of different wireless technologies for IoT deployment on campus with a particular emphasis on studying the energy consumption of different platforms. The focus of this study was on platforms that operate in the sub-GHz band and cellular modules using GSM/GPRS.

In summary, whilst the surveyed studies are focusing on LoRa/LoRaWAN performance in different environments, in this study, we propose to explore the performance of a LoRa-based IoT network deployed in a University campus. We accomplish this through extensive experiments in indoor, outdoor, line-of-sight, non-line-of-sight, obstructed settings.

III. LoRa Overview

LoRa is a communication technology that leverages spread spectrum transmission principles and provides flexibility in terms of choice of a trade-off between range and data rate. The details in this section are summarised from [23] in order to provide a brief technical overview of LoRa. In LoRa modulation, the data to be transmitted modulates a chirp signal with the symbol period of this signal given by,

$$SymPeriod(T) = 2^{SF}/BW(seconds) \tag{1}$$

which results in a symbol rate of

$$SymRate(R) = BW/2^{SF}(symbols/second) \tag{2}$$

The SF values ranging from 7 to 12 imply a range of spreading rates between 2^7 to 2^{12} symbols/second. BW is the bandwidth over which the signal is transmitted taking on one of the values from 125kHz, 250 kHz or 500kHz. LoRa also provides forward error correction capability by adding extra redundant bits in the packet to improve the robustness of transmission. The coding rate is given by,

$$Codingrate = 4/(4 + CR_i) \tag{3}$$

CR_i takes on values between 1 to 4 incrementing in steps of 1 resulting in coding rate values of 4/5, 4/6, 4/7 and

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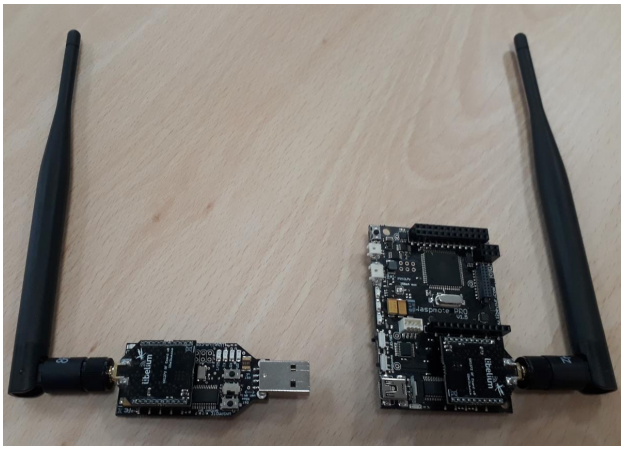

Fig. 1: LoRa platform used in this study

4/8 referred to as CR5, CR6, CR7 and CR8 respectively. The resulting bit rate equation is given by,

$$B = SF * (4/(4 + CR_i))/(2^{SF}/BW)(bits/second) \quad (4)$$

The noise floor is given by

$$Noise = -174 + 10 * log_{10}(B) \quad (5)$$

For a channel Bandwidth of 125kHz, this translates to a figure of $-124dBm$. Most LoRa receivers are able to decode transmissions below the noise floor since LoRa transceivers typically support receiver sensitivities around $-135dBm$. It is worth noting that different combinations of the aforementioned transmission parameters yield different tradeoffs with respect to the range and data rate that can be achieved and these combinations are also governed by regulatory constraints. In terms of deployment, the most popular approach is to use a star topology with LoRa endpoints broadcasting their transmissions which are intercepted by LoRa gateways in their vicinity. The gateways forward these packets to the LoRa network server which handles duplication.

In summary, LoRa is a flexible connectivity solution and quoting from [23] "the use of spread spectrum modulation enables greater transmission distance to be obtained, even in harsh, challenging environnments". In the remainder of this paper, we focus on validating this claim by studying the performance of LoRa through a deployment in the UAE University campus and sharing our findings.

IV. Measurement Methodology

In this study, we used the EU version of the Waspmote LoRa platform (shown in Figure 1) which operates in the 868MHz band. There are eight channels supported by this platform with the distance between two adjacent centre frequencies of 300kHz, the first and the last centre frequencies being 865.20MHz and 868MHz respectively. The node shown in the left half of Figure 1 is the gateway node and on the right, is the end point. For the findings reported in this paper, all tests were conducted using the same hardware and software platform.

The Waspmote LoRa gateway device was connected to a PC located on a desk in a faculty office. The end devices were deployed at different indoor and outdoor locations on the UAEU campus with the aim of covering different types of environments including Line of Sight (LoS) and Non Line of Sight (NLoS) with the latter being affected by obstructions resulting from buildings, trees and multi-path propagation effects. The link measurements at each location were carried out sequentially with each end device and the gateway being configured with identical transmission parameters and cycling through the twenty four combinations in total for the six spreading factor SF values (7, 8, 9, 10, 11, 12) and four coding rate CR values (4/5, 4/6, 4/7 and 4/8)[2] maintaining a constant value of 125 kHz for the bandwith (BW) parameter.

During each of these experiments, each end device was configured to transmit one hundred packets to the LoRa gateway with the gateway configured to respond with an ACK for each of these packets. The standard 4.5dBi omni-directional antenna that ships with the Waspmote LoRa platform was used on both the gateway and the end devices. The externally radiated isotropic power (ERP) configured on all nodes (gateway as well as the end devices) throughout all the tests was fixed at 14dBm, the maximum allowable.

A separate walkabout test was also conducted to assess coverage performance by configuring the end device and gateway with the transmission parameter values of SF 12, CR 5 and BW 125kHz. Using the highest SF value yields the best possible coverage achievable at the cost of greater transmission time on air which increases the likelihood of collisions and therefore, compromises throughput performance.

Before conducting these experiments, an antenna designed to listen to the 868MHz band was connected to a Software Defined Radio (SDR) device and a sniffer program was run to identify any other transmissions ongoing in the different channels of the 868MHz band. No transmissions were detected in this band. Figure 2 shows the FFT and spectrogram of a LoRa transmission originating from one of our LoRa modules transmitting on the 865.20 MHz frequency.

The performance metrics considered in this study were packet delivery ratio (PDR) and the received signal strength indicator (RSSI). The former was calculated as the number of packets successfully delivered to the destination divided by the total number of packets sent with a value of 1 implying 100% success and a value of 0 implying no success at all. The value of RSSI is the one reported by the LoRa gateway device.

Through the above set of experiments, we set out to understand the radio signal strength at the different locations on campus and the impact of different transmission parameters on the PDR in different environments. Given the receiver sensitivity of the chosen platform and the claims in the existing literature of being able to achieve

[2]hereinafter referred to as CR5, CR6, CR7 and CR8

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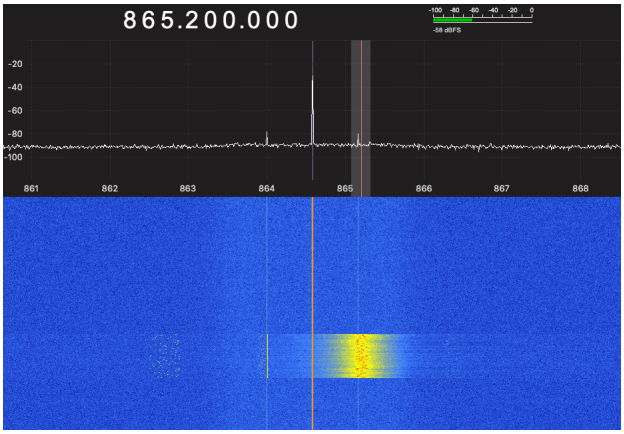

Fig. 2: FFT and spectrogram of a LoRa transmission in the 868MHz band

Fig. 3: Measurement locations (Image Source: [24])

several kms of range in practice, it is also of interest to assess how far in terms of distance could connectivity be established reliably on the campus.

V. Experimental Findings

Figure 3 shows the different locations on the campus where measurements were conducted and Table I shows the approximate straight line distance of the different endpoints from the gateway. Figure 3 also shows the signal strength values (lowest and highest) overlaid on the map for the walkabout test in LoS and NLoS scenarios. During this test, a walk was performed along the dotted lines shown in the figure starting from the building hosting the gateway traversing the locations in the alphabetical order

TABLE I: Node location and distance (approx. straight line) from the gateway GW and RSSI recorded for packets received by the gateway from each of these locations (see Fig-3)

Loc.	Type	Description	Dist.	RSSI(dBm) (max,min)
A	outdoor	LoS location	260m	-101,-119
B	outdoor	location with some obstruction	380m	-120,-125
C	outdoor	heavily obstructed	490m	-134,-135
D	outdoor	some obstructions and possible LoS)	430m	-128,-135
E	outdoor	heavily obstructed	340m	-132,-135
F	outdoor	some obstruction	400m	-124,-128
G	indoor	NLoS location	200m	-127,-136
H	outdoor	obstructed	390m	-128
I	outdoor	obstructed	230m	-127,-131
J	indoor	NLoS	90m	-120,-125
K	indoor	NLoS	80m	-98,-102

and back to the starting point. At each location, the end point was configured to transmit several packets and the lowest and highest RSSI was recorded at the gateway for each of these locations.

As shown in this figure, the worst connectivity was achieved at a distance of 490 metres from the location of the gateway node (location C in Figure 3). Whilst these experiments suggest that optimistically a range of about 490 metres was achievable, in practice, location C is at the edge of the link with vast majority of the transmission attempts failing and only three packets being successfully delivered.

Similar performance was observed at location H which is at a relatively shorter distance from the gateway than the end point at location C. Whilst the observed RSSI at the gateway appears to be better for packets originating from this location in comparison to location C, only two packets could be successfully delivered to the gateway from this location.

Transmissions from locations A, B, D, F and I offered higher success than at other locations because of relatively fewer obstructions unlike the other locations. For example, transmissions from location F had a much better RSSI than location G despite having a longer distance from the gateway. This is attributed to the fact that location G is an indoor location (cafeteria) and therefore the signals have to travel through an obstructed environment unlike those at location F which has little to no obstruction with a long open corridor that leads to the building hosting the gateway.

These results show that with the default 4.5dBi antennas shipped with the Waspmote platform, it is difficult to achieve longer range in an obstructed environment. These findings also indicate that the achievable performance in a practical setting can be much more conservative in comparison to figures reported in the literature, which suggest a range in excess of several kms albeit with a high gain antenna deployed at high rise locations to ensure a line of sight. It is worth noting that unlike some of the other studies that focus on LoS links with gateways

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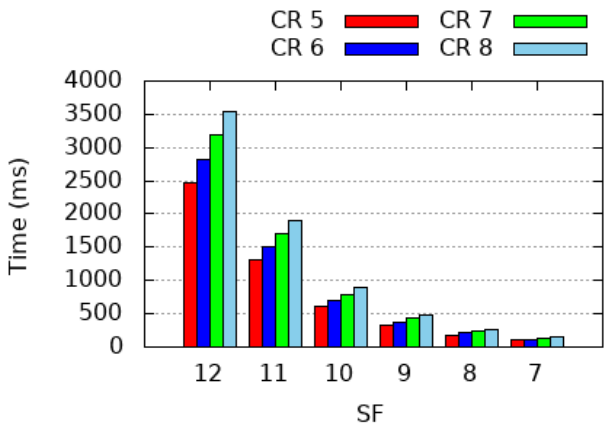

Fig. 4: Time-on-air for different SF and CR values for a 50 byte packet

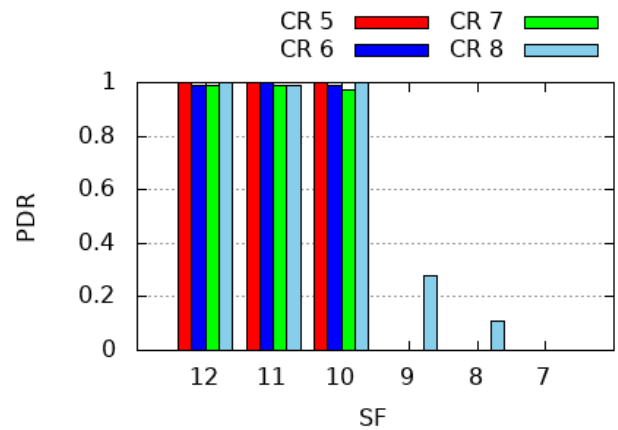

Fig. 5: PDR achieved for transmissions from location E for different SF and CR values

placed at high rise locations, the work described in this paper focuses on non line of sight and indoor, outdoor propagation environment. Therefore, for a campus sensing application deployment, multiple gateways may need to be deployed to provide optimal coverage and performance.

We now turn our attention to detailed link performance measurements for a subset of these locations to understand the performance that could be achieved in practice. Towards this goal, extensive and dedicated link measurements were conducted at the following locations:

- Location E, an outdoor location behind buildings in a heavily obstructed environment
- Location G, an indoor location inside a cafeteria
- Location J, indoor location on the 2nd floor of the same building that hosts the gateway with no LoS of the gateway
- Location K, indoor location on the same floor as the gateway but on the diagonally opposite end of the building with no LoS

A. Time-on-air (ToA) for different transmission parameter settings

Figure 4 shows the Time-on-Air (ToA) for a 50 byte LoRa packet transmitted at different SF and CR values for a fixed bandwidth of 125kHz. As seen from this figure, the higher the SF value for a given CR and BW value, the longer it takes for the packet to be transmitted. Similarly, the higher the redundancy in the packet (CR value) for a given SF and BW value, the longer it takes for packet transmission. It is well-known that the choice of SF and CR provides a compromise between range and data rate. Increasing SF values provide a better range at the cost of a lower achievable data rate. Similarly, increasing CR values (increased redundancy in the packet) offer the ability to decode the packet at relatively longer range at the cost of increasing transmission time and a lower achievable data rate. Whilst this may give one an impression that making such a trade-off decision is trivial, findings from our link measurement study indicate that choosing appropriate

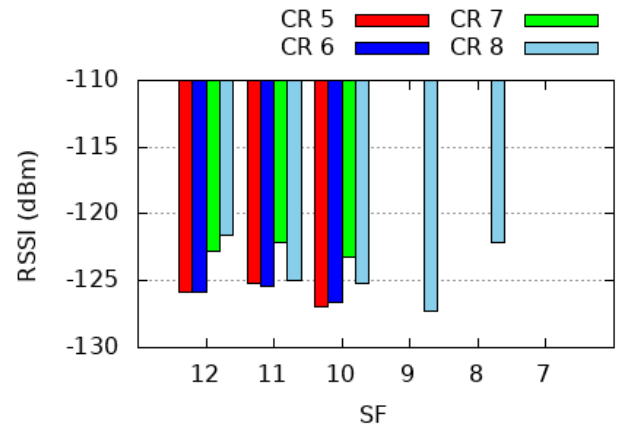

Fig. 6: RSSI registered at the gateway for transmissions from location E for different SF and CR values

values of SF and CR at different deployment locations is far from trivial. Our findings also indicate that sometimes using more conservative SF and/or CR values may not necessarily have the same effect as expected. In particular, we highlight cases which exhibit the law of diminishing returns.

B. Findings from an outdoor obstructed environment

Figure 5 shows the PDR attained for transmissions from location E to the gateway for different combinations of SF and CR values. As evident from this figure, irrespective of the CR value, SF 12, 11 and 10 are well-supported at this location attaining a PDR close to 1. Looking at the graph, it is also clear that there is hardly any need for higher levels of redundancy to support robust communication between the node at this location and the gateway as the PDR attained is close to 1 even for the least level of redundancy (CR=5). For an SF value of 9 and 8, communication was only possible when using the highest level of redundancy (CR=8). Despite this, the connectivity was patchy with only a small fraction of packets being delivered successfully, close to 30% and 10% respectively for the two settings. No communication was possible for

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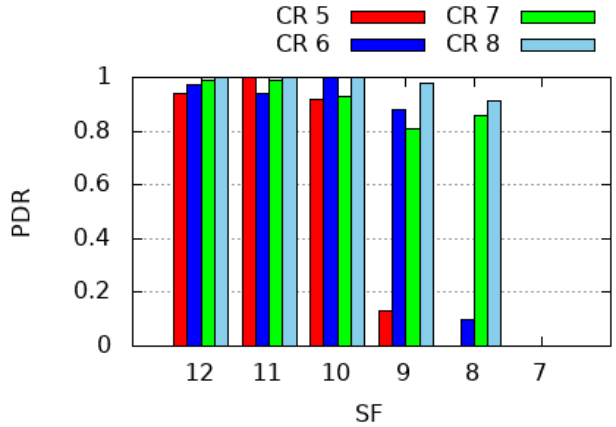

Fig. 7: PDR achieved for transmissions from location G for different SF and CR values

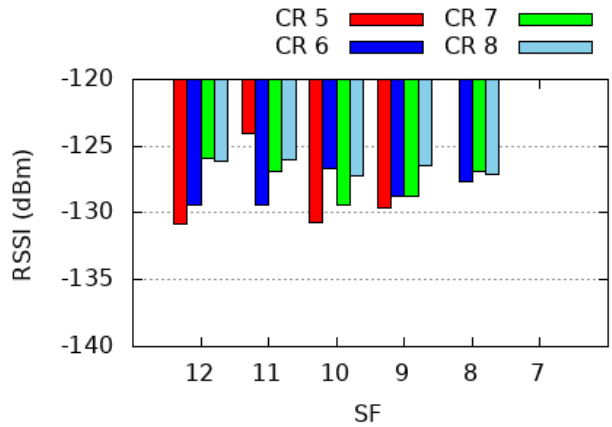

Fig. 8: RSSI registered at the gateway for transmissions from location G for different SF and CR values

an SF value of 7 irrespective of the CR value indicating the unsuitability of this setting.

Figure 6 shows the received signal strength indicator at the gateway for transmissions from location E. As evident from this figure, signal strength values around the -130dBm mark were observed for this location indicating that this link is operating close to its edge[3]. A setting with SF=10 and CR=5 emerges as the best choice for this location to achieve the best possible performance with the least time on air. None the less, the other two alternatives (SF=9 and SF=8) could also be used. However, this would then depend on the application's tolerance to packet loss. If an application does not demand high levels of delivery accuracy and packet loss is tolerable such than 1 in few packets may be successful, then a more aggressive setting could also be used. Therefore, the choice of settings boils down to the demands of the application and what can be practically delivered in the given deployment scenario.

[3]Communication drops off beyond -135dBm

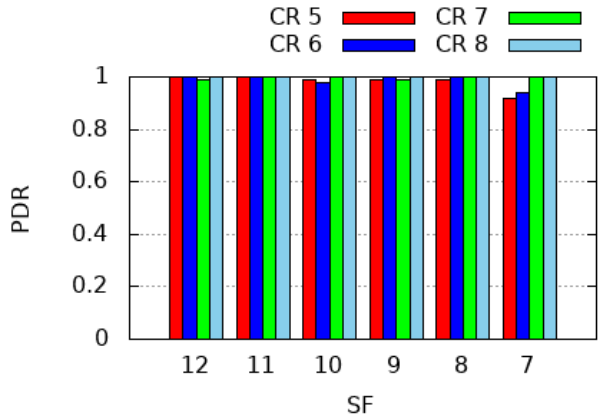

Fig. 9: PDR achieved for transmissions from location J for different SF and CR values

C. Findings from a mixed outdoor-indoor environment

Figure 7 shows the PDR attained for transmissions from location G to the gateway for different combinations of SF and CR values. As mentioned earlier, location G is an indoor obstructed location (inside a student cafeteria) with no LoS of the gateway. This is a relatively more dynamic environment with human mobility impacting the signal propagation in addition to the losses resulting from signal penetration through buildings. Also, this location is relatively closer in terms of physical proximity to the gateway than location E. Observe from Figure 7 that communication from this location to the gateway is possible for a variety of parameter settings. For SF values up to 10, almost all settings yield a high PDR close to 1. For an SF value of 9 and the default CR value of 5 in this setting, a low PDR is achieved (just over 10%) whereas for the same setting with an SF value of 8, no communication is possible as not a single packet is delivered. As the redundancy level in the packet is increased (CR 6), the PDR improves for the SF 9 setting. A similar change for SF 8 setting yields an improvement in terms of connectivity being enabled albeit with a low PDR performance around 10% suggesting that an even higher CR is required to further improve performance for this SF setting. As the CR is further increased to CR 7 and 8, dramatic improvement in performance is observed indicating that to successfully decode packets for such aggressive SF values, higher levels of CR are essential. Similar to the previous scenario, no communication was achieved for SF value of 7.

Figure 8 shows the RSSI observed at the gateway for transmissions originating from this location. As evident from this figure, this link is also operating close to the edge. Findings from these experiments indicate that with the hardware and antenna used in this study, it is possible to achieve communication in indoor obstructed environments at a severely curtailed range despite the fact that these transmissions are occurring in the sub-GHz band which are known to have better signal propagation characteristics.

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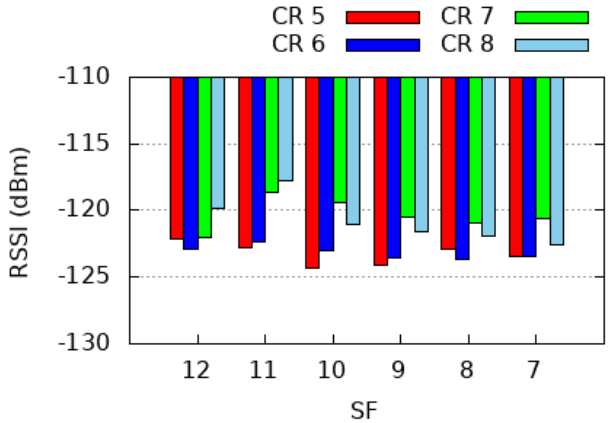

Fig. 10: RSSI registered at the gateway for transmissions from location J for different SF and CR values

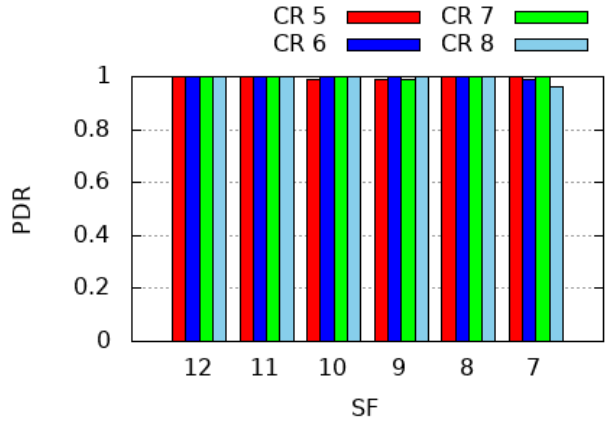

Fig. 11: PDR achieved for transmissions from location K for different SF and CR values

D. Findings from an indoor obstructed environment

Figure 9 shows the PDR attained for transmissions from location J to the gateway for different combinations of SF and CR values. This location is on a floor below the location of the gateway in the same building but in the symmetrically opposite wing of the building at a distance of around 90m from the gateway location. The signals from this location have to penetrate through the building in order to reach the gateway. The end device was placed on a desk close to a big glass window and therefore, existence of a multi-path link to the gateway cannot be ruled out. Unlike the other locations considered so far, a stable performance was observed in this setup. As evident from Figure 9, a high PDR is observed for all combinations of SF and CR values suggesting that the most aggressive parameter setting that yields the highest data rate and lowest time on air can be easily supported at this location. This is attributed to the closer physical proximity between the gateway and end point location and a plausible existence of a multi-path link. Figure 10 shows the RSSI observed at the gateway for transmissions originating from location J. As seen from this figure, a strong signal is registered by the gateway with the average RSSI around the -120dBm mark.

Finally, Figure 11 shows the PDR attained for transmissions from location K to the gateway for different combinations of SF and CR values. This location is on the same floor as the gateway but in the diagonally opposite wing of the building at a distance of around 80m from the gateway location. Similar to the previous setting, signals from this location have to penetrate through the building to reach the gateway and there is also a possible multi-path link given signal deflections from the outer glass walls of the building. Similar to that in the previous scenario, a stable performance was observed in this setup with a PDR close to 1 achievable for all combinations of SF and CR values as seen from Figure 11. This is also confirmed by the RSSI plot (12) which shows that a strong signal (varying between -100dBm to -120dBm) was registered

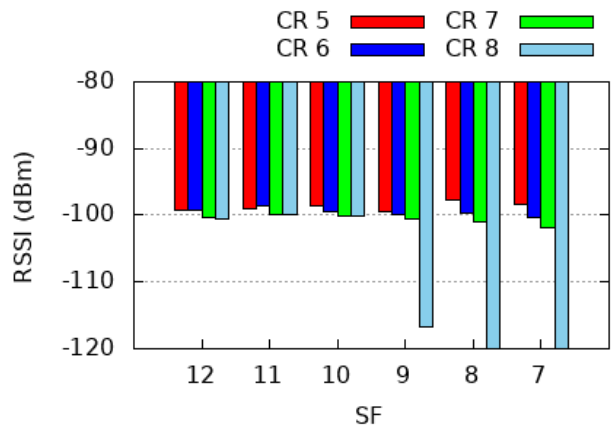

Fig. 12: RSSI registered at the gateway for transmissions from location K for different SF and CR values

at the gateway for transmissions originating from this location. This implies that the highest possible data rate supported by LoRa (for the SF 7 CR 5 setting) can be achieved for this setting implying the lowest time on air for the transmissions.

From the tests carried out in this study, it was found that at best a range of around 500m was achievable in an obstructed NLoS environment and that too for a high transmit power setting. These numbers are more conservative than the claims made so far which prompt caution when taking numbers at face value. Having said this, one must note that this work deals with a lot of NLoS and indoor propagation links, whereas other research typically describes more LoS links between well-placed antennas. Also, the choice of conservative parameter settings, e.g. using worse SF and CR values depending on the deployment scenario, have performance implications. Whilst the range and performance may be compromised, this can be made up for with deployment of multiple gateways. As has been shown in our earlier work [25], there are plethora of connectivity technology alternatives that can be leveraged for any IoT deployment each having their own pros and cons. However, the cost vs benefit tradeoff needs to be

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evaluated to ensure that the choice meets the operational and commercial requirements.

VI. Discussion

There is no doubt that LPWAN technologies bridge the gap between the short range wireless and the cellular technology alternatives. Therefore, one cannot help but notice the advantages of LPWAN technologies in general and LoRa in particular given its open specification, ready availability of commercial-off-the-shelf hardware and software and a strong industry alliance driving the push for adoption. Having said this, several technology and commercial challenges remain to be resolved. It is clear that the performance achievable with LoRa can vary greatly depending on the deployment scenario and is sensitive to the choice of operational parameters. Whilst there are conditions under which choice of conservative parameter values, e.g. using worse SF and CR values can mean the difference between connectivity and the lack thereof, there are other situations where aggressive parameter values can be safely leveraged to maximise performance. Identifying the optimal value for these operational parameters and adapting them in response to changes in the operational environment remains a challenge.

Secondly, what remains to be seen is the impact of choices made by individual nodes on the overall performance. It is well known that choosing a lower modulation coding scheme (MCS) results in slower transmissions and choosing a higher MCS results in relatively quicker transmissions. If a node chooses lower MCS values, it ends up occupying the medium for a longer time thereby increasing the likelihood of collisions especially if there are a lot of other nodes competing for transmissions at the same time. This problem, known as rate poisoning, can drag the network performance down. It has been shown in [26] how this problem affects the performance of Wireless LANs. Ideally, it is desirable for nodes to use higher MCS values where possible so that they do not have a negative impact on network performance (avoid rate poisoning). It is not hard to imagine that such effects might come to bear on LoRa based networks too (nodes choosing the worst SF, CR values), a problem that needs to be investigated. Whilst LoRa based networks were designed to operate in the unlicensed sub-GHz spectrum, coexistence challenges exist especially for other solutions leveraging the same bands. Severe duty cycling and operating amidst interference from other networks / technologies using these spectrum bands will continue to be a challenge which has prompted network operators to ponder over whether deploying this in the licensed band makes sense. However, the technical and commercial issues surrounding this remain open.

VII. Conclusions

In this article, we set out to investigate the performance of LoRa, a long range sensor networking technology, through a deployment in a University campus. This was accomplished by carrying out extensive performance measurements for a variety of transmission parameter settings by deploying end points at different indoor and outdoor locations covering line of sight and non line of sight scenarios affected by obstructions and multipath propagation effects. Findings from this study highlight that the achievable performance can vary greatly depending on the deployment scenario and parameter configuration. Whilst our findings concur with prior studies that LoRa performance is sensitive to parameter settings, we found that the range in practice in an obstructed environment can be significantly compromised and in the range of several hundreds of metres unlike the kms of range reported in existing literature which prompts caution when taking numbers at face value. It should however be noted that this work deals with a lot of NLoS and indoor propagation links, whereas other research typically describes more LoS links between well-placed antennas. Our findings also highlight the law of diminishing returns resulting from incremental performance improvements through inclusion of increased redundancy in the packet and a consequent increase in time-on-air which may have other potential detrimental effects such as an increased risk of collisions that will impact network performance. Whilst the range and performance may be compromised in a NLoS/indoor propagation scenario, this can be made up for with deployment of multiple gateways which will drive up the cost and complexity. Future work could look into the the cost vs benefit trade-off of such an approach to meet operational and commercial requirements.

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